Shadrak BESSANH

CONTACT

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PROFILE

Passionate Embedded Systems & IoT Engineer with hands-on experience designing and building intelligent energy systems. Specialized in combining IoT technologies with data analytics and control systems for renewable energy applications.

EDUCATION

- BACHELOR'S DEGREE IN EMBEDDED SYSTEMS AND IOT

  2023 - Present

  INSTITUTE FOR TRAINING AND RESEARCH IN COMPUTER SCIENCE

- SCIENTIFIC BACCALAUREATE

  2023

  COLLEGE CATHOLIQUE D'AZOVE

SKILLS

- ESP32 & Arduino microcontroller programming
- Sensor integration & data acquisition
- Real-time control systems & automation
- Firmware development & embedded C/C++
- IoT device communication & networking
- Python (scripting, automation, data analysis)
- API design & integration (RESTful, MQTT)

LANGUAGES

- French: Native
- English: Professional Proficiency

PROJECT EXPERIENCE

BACKEND DEVELOPER

2023-2025

NETCOP SARL

- Developed and maintained RESTful APIs for web applications
- Designed and optimized MySQL databases
- Implemented secure authentication systems
- Participated in code reviews and technical documentation
- Collaborated with frontend team for feature integration

FEATURED PROJECTS

PREDICTIVE ANALYTICS FOR SUSTAINABLE ENERGY

- Designed IoT system to collect energy consumption data from multiple sources
- Built Python-based predictive model analyzing historical energy patterns
- Implemented real-time monitoring dashboard for energy usage optimization
- Integrated sensor data with backend analytics pipeline
- Tech: ESP32, MQTT, Python, SQL, data visualization

RENEWABLE ENERGY FORECASTING SYSTEM

- Analyzed historical energy generation data to predict renewable output
- Built Python scripts for data cleaning, transformation, and analysis
- Designed database schema for time-series energy data
- Created visualization dashboards for stakeholder reporting
- Tech: Python, Pandas, SQL, data analytics
- Impact: Improved planning for renewable energy deployment